

An explicit 2^{3200} commutator bound via simultaneous block elimination

Informal proof with a Lean correspondence appendix

September 17, 2026

Revision following the supplied referee report

Abstract

Every trace-zero complex matrix A admits a representation $A = BC - CB$ in its original dimension with $\|B\| \|C\| \leq 2^{3200} \|A\|$, where all norms are Euclidean operator norms. Following the supplied referee report, two simultaneous square-zero shears replace the successive outside-block eliminations. This removes the iterated polynomial from the high trace-mass branch. The trace-preserving low-mass paving and the same-dimension strong induction are retained. We give the full informal commutator proof, the numerical estimates leading to 2^{3200} , and a correspondence to the completed Lean development.

Scope of this revision. This document revises the commutator proof in the repository’s `paper/main.tex`, using Comment 3 of the supplied report dated 15 September 2026. The two-shear improvement is credited to that report. The other mathematical ingredients are retained from the existing manuscript and its formalization. The separation-family theorem and the report’s optional corollaries are outside this document’s scope.

Formal result. The new declaration `NoEpsilon.commutator_bound_two_pow_3200` has the explicit constant in its conclusion and no additional branch hypotheses. Its factors act on the same $\mathbb{C}^n$ as A ; Appendix A records norm semantics, dependencies, and reproducible checks.

Contents

1	Introduction and main theorem	2
2	An elementary two-commutator decomposition	3
3	Solving an identity-corner core	3
4	Absorbing a bounded number of remaining blocks	5
5	The high trace-mass branch	6
6	The low trace-mass branch	8
7	Finite-block assembly and the global induction	11
A	Lean correspondence and reproducibility	13

1 Introduction and main theorem

For $A \in M_n(\mathbb{C})$, write $\|A\|$ for its operator norm on Euclidean space and $[B, C] = BC - CB$. The quantitative commutator problem asks how economically one can represent a trace-zero matrix by a single commutator. Johnson, Ozawa, and Schechtman proved that, for each $\varepsilon > 0$, a bound $K_\varepsilon n^\varepsilon \|A\|$ is possible; their first factor can be chosen normal [3]. Ravichandran and Srivastava obtained the bound $300 \exp(9\sqrt{\log n}) \|A\|$ [6, Corollary 2]. The theorem considered here allows both factors to be arbitrary complex matrices.

Theorem 1.1 (Main theorem). *With $K = 2^{3200}$, for every integer $n \geq 1$ and every $A \in M_n(\mathbb{C})$ with $\text{tr } A = 0$, there exist $B, C \in M_n(\mathbb{C})$ satisfying*

$$A = [B, C], \quad \|B\| \|C\| \leq K \|A\|. \quad (1.1)$$

The same K works for every n .

The construction has two branches. Set

$$s = 8192, \quad R = s^2 = 2^{26}, \quad b = 8sR = 2^{42}, \quad t_0 = 2/R. \quad (1.2)$$

A matrix with large trace norm in every rotated Hermitian direction is handled directly, with a constant $K_H(t_0)$. Otherwise it admits R trace-zero compressions of norm at most $319/R$ times its own norm. Reassembly costs a factor $8s$ on the largest child budget, and

$$8s \frac{319}{R} = \frac{2552}{8192} < \frac{1}{2}.$$

This strict inequality is the quantitative reason the induction closes.

1.1 Conventions and elementary norm estimates

All spaces are finite-dimensional complex Hilbert spaces. A compression to a subspace V means $P_V A|_V$; no invariance of V is implicit. Rectangular blocks carry their operator norms. The trace is unnormalized, and $|H| = (H^2)^{1/2}$ for Hermitian H . We use

$$\text{Re } A = (A + A^*)/2, \quad \text{Im } A = (A - A^*)/(2i), \quad \chi(S) = \|S\| \|S^{-1}\|.$$

Unitary conjugation preserves the norm and trace. A similarity multiplies a commutator product budget by at most $\chi(S)^2$.

We shall use a basic block estimate repeatedly. If an operator $T = (T_{ij})$ has zero diagonal in an orthogonal decomposition with k blocks and $\|T_{ij}\| \leq c$ for $i \neq j$, then

$$\|T\| \leq c(k-1). \quad (1.3)$$

Indeed, for $x = (x_j)$, the vector of component norms of Tx is bounded coordinatewise by $c(\mathbf{1}\mathbf{1}^\top - I)(\|x_j\|)_j$. The scalar matrix has Euclidean operator norm $k-1$ when $k \geq 2$; the case $k=1$ is immediate. Also, the norm of a block diagonal operator is the largest block norm, and the norm of a general block matrix is at most the sum of its block norms.

1.2 External inputs

Besides finite-dimensional spectral theory, the Toeplitz–Hausdorff theorem, compact convexity, and the Banach fixed-point theorem, the proof uses three specific published inputs: the two-dimensional Steinitz rearrangement theorem, the simultaneous hollowisation theorem of Damm and Faßbender, and the independent-vector theorem of Marcus, Spielman, and Srivastava. Their exact forms are stated in Section 6. The two-commutator decomposition in the next section is elementary and allows its first factors to depend on the input matrix.

2 An elementary two-commutator decomposition

Lemma 2.1. *Let $m \geq 1$. If $G \in M_m(\mathbb{C})$ is Hermitian and $\operatorname{tr} G = 0$, there are a unitary u and a matrix V such that $G = [u, V]$ and $\|V\| \leq \|G\|$.*

Proof. For $m = 1$, take $u = I$ and $V = 0$. Otherwise diagonalize G unitarily and denote its real eigenvalues by $\lambda_1, \dots, \lambda_m$. They sum to zero and lie in $[-a, a]$, where $a = \|G\|$. They can be ordered so that every partial sum $s_j = \sum_{i=1}^j \lambda_i$ belongs to $[-a, a]$. To see this, if the current sum is positive, choose a remaining negative term; such a term exists because the remaining terms have negative sum. If the current sum is negative, choose a positive term. At zero choose any remaining term. In each case the new sum stays in $[-a, a]$. This constructs the required ordering, including zero terms.

In this ordered eigenbasis, let

$$u = \sum_{j=1}^{m-1} E_{j,j+1} + E_{m,1}, \quad V = \sum_{j=1}^{m-1} s_j E_{j+1,j}.$$

The first matrix is unitary, $\|V\| = \max_{j < m} |s_j| \leq a$, and multiplication gives

$$uV = \operatorname{diag}(s_1, \dots, s_{m-1}, 0), \quad Vu = \operatorname{diag}(0, s_1, \dots, s_{m-1}).$$

Their difference is $\operatorname{diag}(\lambda_1, \dots, \lambda_m)$, since $s_m = 0$. Conjugate both factors back to the original basis. $\square$

Corollary 2.2. *Let $m \geq 1$. Every trace-zero $H \in M_m(\mathbb{C})$ has a representation*

$$H = [u, V] + [k, T], \quad u, k \text{ unitary}, \quad \max(\|V\|, \|T\|) \leq \|H\|. \quad (2.1)$$

Proof. Write $H = H_1 + iH_2$ with H_1, H_2 Hermitian, trace zero, and of norm at most $\|H\|$. Apply Lemma 2.1 to obtain $H_1 = [u, V]$ and $H_2 = [k, W]$, and set $T = iW$. $\square$

Remark 2.3. The quantifiers in (2.1) are $\forall H \exists u, k, V, T$. We do not assert that one pair u, k works uniformly for all H in a fixed dimension, or that the factors vary continuously with H . Neither property is needed below: the four factors are chosen once before the Riccati equation is solved.

3 Solving an identity-corner core

For $a \geq 1$, define the following scalar quantities:

$$\begin{aligned} p &= 2a, & d_0 &= 5a, & q &= 5a + 1, & d_1 &= 14a + 1, \\ L_0 &= q^2 + (2a + 1)q + d_1, & L_1 &= 2q + 2a + 1, & \lambda &= 2 \max(L_0, L_1), \\ P(a) &= (1 + q)^4(\lambda + 3)((\lambda + 5)p + 2). \end{aligned} \quad (3.1)$$

All of these are finite and nondecreasing in a .

Proposition 3.1. *Let $m \geq 1$, $a \geq 1$, and*

$$M = \begin{pmatrix} A & I_m \\ F & D \end{pmatrix}, \quad \text{tr } M = 0, \quad \|M\| \leq a.$$

Then M is a single commutator in $M_{2m}(\mathbb{C})$ with factor norm product at most $P(a)$.

Proof. Since $\text{tr}(A + D) = 0$, Corollary 2.2 supplies fixed factors such that

$$A + D = [u, V] + [k, T], \quad \|u\| = \|k\| = 1, \quad \|V\|, \|T\| \leq p.$$

Set

$$U = \lambda I + u, \quad D_0 = [u, V] - A, \quad D_1 = [k, V] + D_0 u - k + [u, Tk] - F. \quad (3.2)$$

The block compression bounds imply $\|A\|, \|D\|, \|F\| \leq a$, whence

$$\|D_0\| \leq 2p + a = d_0, \quad \|D_1\| \leq 2p + d_0 + 1 + 2p + a = d_1.$$

On the closed ball $\mathcal{B} = \{Q : \|Q - D_0\| \leq 1\}$ consider

$$\Psi(Q) = D_0 + \lambda^{-1}(Q^2 - DQ + Q(A - u) + D_1). \quad (3.3)$$

Every $Q \in \mathcal{B}$ has norm at most q , so

$$\|\Psi(Q) - D_0\| \leq L_0/\lambda \leq \frac{1}{2}.$$

For $Q, Q' \in \mathcal{B}$, the noncommutative identity $Q^2 - Q'^2 = (Q - Q')Q + Q'(Q - Q')$ gives

$$\|\Psi(Q) - \Psi(Q')\| \leq \frac{2q + 2a + 1}{\lambda} \|Q - Q'\| \leq \frac{1}{2} \|Q - Q'\|.$$

The ball is complete. The Banach fixed-point theorem gives an exact fixed point $Q \in \mathcal{B}$.

Define

$$B_c = \begin{pmatrix} U & I \\ k & 0 \end{pmatrix}, \quad C_c = \begin{pmatrix} V & T \\ A + Q - [u, V] + Tk & V + I - UT \end{pmatrix}, \quad S = \begin{pmatrix} I & 0 \\ Q & I \end{pmatrix}. \quad (3.4)$$

To verify the order of all factors, put

$$F_0 = [k, V] + D_0 U - k + [u, Tk] = \lambda D_0 + D_1 + F.$$

Direct multiplication, using $A + D = [u, V] + [k, T]$, yields

$$\begin{aligned} [B_c, C_c] &= \begin{pmatrix} A + Q & I \\ F_0 - QU & D - Q \end{pmatrix}, \\ S^{-1}MS &= \begin{pmatrix} A + Q & I \\ F + DQ - QA - Q^2 & D - Q \end{pmatrix}. \end{aligned} \quad (3.5)$$

The bottom-left blocks agree precisely when

$$Q^2 - DQ + Q(A - U) + F_0 - F = 0,$$

which is equivalent to (3.3). Consequently

$$M = [SB_c S^{-1}, SC_c S^{-1}]. \quad (3.6)$$

It remains to bound the factors. The block norm sum gives $\|B_c\| \leq \lambda + 3$. The bottom-left block of C_c equals $Q - D_0 + Tk$, of norm at most $1 + p$, and the bottom-right block has norm at most $1 + (\lambda + 2)p$. Thus

$$\|C_c\| \leq (\lambda + 5)p + 2, \quad \|S\|, \|S^{-1}\| \leq 1 + q.$$

Equation (3.6) now gives the claimed $P(a)$. $\square$

4 Absorbing a bounded number of remaining blocks

Lemma 4.1 (Sylvester equation with separated centers). *Let U and V act on possibly different Hilbert spaces, and let $z, w \in \mathbb{C}$. If $\eta = \|U\| + \|V\| < |z - w|$, then, for every rectangular Y , the equation*

$$(z - w)X + UX - XV = Y$$

has a solution with $\|X\| \leq \|Y\| / (|z - w| - \eta)$.

Proof. On the space of rectangular operators with its operator norm, the map $\mathcal{E}(X) = UX - XV$ has norm at most η . The inverse of $(z - w)I + \mathcal{E}$ is the norm-convergent Neumann series

$$\frac{1}{z - w} \sum_{j=0}^{\infty} \left(-\frac{\mathcal{E}}{z - w} \right)^j,$$

whose norm is at most $(|z - w| - \eta)^{-1}$. □

Proposition 4.2. *Let a trace-zero matrix A have an orthogonal block decomposition of sizes $r, r, s_3, \dots, s_b$, where $b \geq 2$, $r \geq 1$, and $1 \leq s_j \leq r$. Suppose $A_{12} = I_r$ and $\|A\| \leq a$, with $a \geq 1$. Put*

$$\begin{aligned} q &= b - 2, & u &= 1 + a + \sqrt{q}, & a_1 &= au^2, \\ v &= 1 + \sqrt{q}a_1, & \hat{a} &= a_1v^2, & k &= q + 1. \end{aligned} \tag{4.1}$$

Then A has a same-dimensional single-commutator representation of cost at most

$$(\hat{a}/a)^2 k (4P(\hat{a}) + 2k\hat{a}). \tag{4.2}$$

Proof. Let $F = \bigoplus_{j=3}^b \mathbb{C}^{s_j}$. Choose isometries $J_j : \mathbb{C}^{s_j} \rightarrow \mathbb{C}^r$ and form the row operator $J = (J_3 \ \cdots \ J_b) : F \rightarrow \mathbb{C}^r$. Its summands need not have orthogonal ranges. Nevertheless,

$$JJ^* = \sum_{j=3}^b J_j J_j^* \leq qI_r, \quad \|J\| \leq \sqrt{q}.$$

Use three large blocks of sizes $r, r, \dim F$, and put

$$X = J - A_{1F}, \quad S_1 = \begin{pmatrix} I & 0 & 0 \\ 0 & I & X \\ 0 & 0 & I \end{pmatrix}, \quad A' = S_1^{-1} A S_1.$$

The off-diagonal part of S_1 squares to zero, so changing the sign of X gives its inverse. Left multiplication leaves the first block row fixed, and right multiplication therefore gives

$$A'_{1F} = A_{1F} + A_{12}X = J, \quad A'_{12} = I_r.$$

In particular, $A'_{1j} = J_j$ for every outside block j .

Now form a column operator $Y : \mathbb{C}^r \rightarrow F$ with block rows $Y_j = A'_{jj}J_j^*$, and set

$$S_2 = \begin{pmatrix} I & 0 & 0 \\ 0 & I & 0 \\ Y & 0 & I \end{pmatrix}, \quad \hat{A} = S_2^{-1} A' S_2.$$

Again the inverse is obtained by changing the sign of the off-diagonal part. Right multiplication changes only block column 1. Left multiplication subtracts Y_j times the first block row from each outside row j . Thus,

$$\hat{A}_{jj} = A'_{jj} - Y_j A'_{1j} = A'_{jj} - A'_{jj} J_j^* J_j = 0 \quad (j \geq 3), \quad \hat{A}_{12} = I_r. \quad (4.3)$$

These identities account for interactions between distinct outside blocks: no successive elimination or disjointness of the ranges of J_j is assumed.

The rectangular compression satisfies $\|A_{1F}\| \leq a$, so

$$\|X\| \leq a + \sqrt{q}, \quad \|S_1\|, \|S_1^{-1}\| \leq u, \quad \|A'\| \leq a_1.$$

For $z \in \mathbb{C}^r$ the orthogonal sum defining F gives

$$\|Yz\|^2 = \sum_{j=3}^b \|A'_{jj} J_j^* z\|^2 \leq q a_1^2 \|z\|^2.$$

Hence $\|Y\| \leq \sqrt{q} a_1$ and $\|S_2\|, \|S_2^{-1}\| \leq v$. Taking $S = S_1 S_2$ yields

$$\hat{A} = S^{-1} A S, \quad \|\hat{A}\| \leq a u^2 v^2 = \hat{a}, \quad \chi(S) \leq u^2 v^2 = \hat{a}/a. \quad (4.4)$$

If $q = 0$, both off-diagonal operators are zero and both shears are the identity; the displayed upper bounds remain valid. Empty outside slots may also be included, which is convenient for the formal fixed-slot partition.

Merge the first two blocks into one core M . Since every other diagonal block is zero and similarity preserves trace, $\text{tr } M = 0$. Proposition 3.1 gives $M = [B_0, C_0]$ with product at most $P(\hat{a})$. Its identity corner ensures $M \neq 0$. Rescale the two factors inversely to arrange

$$\|B_0\| = 1/4, \quad \|C_0\| \leq 4P(\hat{a}).$$

There are now k blocks. Let

$$\hat{B} = \text{diag}(B_0, I, 2I, \dots, (k-1)I), \quad \hat{C}_{11} = C_0, \quad \hat{C}_{jj} = 0 \quad (j > 1).$$

For distinct blocks solve $\hat{B}_{ii} \hat{C}_{ij} - \hat{C}_{ij} \hat{B}_{jj} = \hat{A}_{ij}$ using Lemma 4.1. The centers are distinct integers and the only non-scalar perturbation has norm $1/4$. Thus $\|\hat{C}_{ij}\| \leq 2\hat{a}$ and

$$\|\hat{B}\| \leq k, \quad \|\hat{C}\| \leq 4P(\hat{a}) + 2(k-1)\hat{a}$$

by (1.3). These choices give $[\hat{B}, \hat{C}] = \hat{A}$. Conjugating back and applying (4.4) proves (4.2). $\square$

5 The high trace-mass branch

For $0 < t \leq 1$, say that $A \in M_n(\mathbb{C})$ satisfies $\text{HM}(t)$ if

$$\text{tr} |\text{Re}(e^{i\theta} A)| \geq tn \|A\| \quad \text{for every } \theta \in \mathbb{R}. \quad (5.1)$$

Lemma 5.1 (A disk in compressed numerical ranges). *Suppose $\text{tr } A = 0$, $\|A\| = 1$, and A satisfies $\text{HM}(t)$. If W has codimension $d \leq tn/4$, then the numerical range of $T = P_W A|_W$ contains the closed disk of radius $\rho = t/4$ centered at zero.*

Proof. For every θ , the Hermitian matrix $H_\theta = \operatorname{Re}(e^{i\theta}A)$ has norm at most one and trace zero. Its positive eigenvalues therefore have sum at least $tn/2$. Write the eigenvalues in decreasing order. Since $d < tn/2$, at least $d+1$ eigenvalues are positive; moreover

$$tn/2 \leq d + (n-d)\lambda_{d+1}(H_\theta).$$

It follows that

$$\lambda_{d+1}(H_\theta) \geq \frac{tn/2 - d}{n - d} \geq t/4.$$

The min–max principle yields $\lambda_{\max}(P_W H_\theta|_W) \geq t/4$. The numerical range is compact and convex by the Toeplitz–Hausdorff theorem. Its support in each direction is this largest eigenvalue, so it contains the stated disk. Equivalently, use the supporting-half-plane description of the ordinary numerical range; see [4, Section 2]. $\square$

Lemma 5.2 (A neutral vector with a large image). *If the numerical range of T contains both ρ and $-\rho$, where $\rho > 0$, then there is a unit vector v such that*

$$v^* T v = 0, \quad \|T v\| \geq \rho. \quad (5.2)$$

Proof. Choose unit vectors x_+, x_- with expectations $\rho, -\rho$. The density matrix $D_* = (x_+ x_+^* + x_- x_-^*)/2$ has trace one, satisfies $\operatorname{tr}(T D_*) = 0$, and obeys $\operatorname{tr}(T^* T D_*) \geq \rho^2$ by Cauchy–Schwarz.

On the compact convex set

$$\mathcal{D} = \{D \geq 0 : \operatorname{tr} D = 1, \operatorname{tr}((\operatorname{Re} T)D) = 0, \operatorname{tr}((\operatorname{Im} T)D) = 0\},$$

maximize the real linear functional $\operatorname{tr}(T^* T D)$. The set of maximizers is a nonempty compact face, so it has an extreme point that is also extreme in $\mathcal{D}$. Every extreme point of $\mathcal{D}$ has rank one. In fact, if an extreme D had rank $s \geq 2$, the Hermitian operators on its support would form a real vector space of dimension $s^2 \geq 4$. The three displayed real linear constraints admit a nonzero supported Hermitian perturbation E in their common kernel. Because D is strictly positive on its support, both $D + \eta E$ and $D - \eta E$ belong to $\mathcal{D}$ for sufficiently small $\eta > 0$, a contradiction. The maximizing extreme point is thus vv^* , and its objective value is at least that of D_* . $\square$

Proposition 5.3. *Every trace-zero matrix satisfying $\operatorname{HM}(t)$ has a same-dimensional single commutator representation with factor product at most $K_H(t) \|A\|$, where*

$$\begin{aligned} L &= \lceil 16/t \rceil, \quad a_0 = 4/t, \quad u_* = 1 + a_0 + \sqrt{L}, \\ a_{1,*} &= a_0 u_*^2, \quad a_* = a_{1,*}(1 + \sqrt{L} a_{1,*})^2, \quad k_* = L + 1, \\ K_H(t) &= a_*^2 k_* (4P(a_*) + 2k_* a_*) < \infty. \end{aligned} \quad (5.3)$$

Proof. The zero matrix is immediate. Normalize $\|A\| = 1$, and set $\rho = t/4$ and $r = \lceil tn/16 \rceil$. We construct orthonormal $v_1, \dots, v_r$ such that all $v_i^* A v_j$ vanish and the vectors $A v_i$ are pairwise orthogonal with norms at least ρ .

After selecting $v_1, \dots, v_j$, work in

$$W_j = \operatorname{span}\{v_i, A v_i, A^* v_i, A^* A v_i : 1 \leq i \leq j\}^\perp.$$

For $0 \leq j \leq r-1$ we have $\operatorname{codim} W_j \leq 4j < tn/4$. Lemmas 5.1 and 5.2, applied to $P_{W_j} A|_{W_j}$, give a unit $v_{j+1} \in W_j$ with

$$v_{j+1}^* A v_{j+1} = 0, \quad \|A v_{j+1}\| \geq \|P_{W_j} A v_{j+1}\| \geq \rho.$$

Orthogonality to v_i preserves orthonormality; orthogonality to Av_i and A^*v_i makes both mixed matrix coefficients vanish; orthogonality to A^*Av_i makes Av_{j+1} perpendicular to Av_i . This proves the induction.

Let $V = \text{span}\{v_i\}$ and $W = A(V)$. The construction implies $V \perp W$ and $\dim V = \dim W = r$; in particular $2r \leq n$. In the ordered bases $w_i = Av_i / \|Av_i\|$ of W , followed by v_i of V , the $(1, 2)$ block of A is

$$D = \text{diag}(\|Av_1\|, \dots, \|Av_r\|), \quad \rho I \leq D \leq I.$$

The similarity $S_0 = \text{diag}(I_r, D^{-1}, I_{\text{rest}})$ has condition number at most $4/t$. In $S_0^{-1}AS_0$, the $(1, 2)$ block equals I_r and the norm is at most $a_0 = 4/t$.

Partition the remaining space into blocks of positive size at most r . Since $r \geq tn/16$, the number of outside blocks is at most $L = \lceil 16/t \rceil$. Apply Proposition 4.2. Every factor in its new budget is nondecreasing in the nonnegative outside-block count, so the norm bound is at most a_* , and after including S_0 the total similarity condition number is at most $(4/t)(a_*/a_0) = a_*$. There are at most k_* blocks after merging the core. Its proof gives precisely the bound (5.3). Finally undo the scalar normalization. $\square$

6 The low trace-mass branch

The key point of this branch is a complete paving whose individual traces are exactly zero and whose ranks are equal.

6.1 Rearrangement, hollowisation, and vector selection

We use the following form of the Steinitz theorem: if $u_1, \dots, u_N \in \mathbb{R}^2$ sum to zero and $\|u_i\|_\infty \leq 1$, they can be reordered so that every partial sum has ℓ_∞ norm at most 2. This is the $d = 2$ instance of [1, Theorem 1.2].

Damm and Faßbender prove that three trace-zero Hermitian matrices can be simultaneously conjugated by a unitary so that the first two have zero diagonal and the third has at most two nonzero diagonal entries [2, Proposition 13(b)]. The last two entries need not vanish.

Lemma 6.1. *Let H, G be Hermitian operators on $\mathbb{C}^R$, with $R \geq 2$, and let $E \geq 0$. There is an orthonormal basis $v_1, \dots, v_R$ such that*

$$v_a^* H v_a = \frac{\text{tr } H}{R}, \quad v_a^* G v_a = \frac{\text{tr } G}{R}, \quad v_a^* E v_a \leq \frac{2 \text{tr } E}{R} \quad (1 \leq a \leq R). \quad (6.1)$$

Proof. Subtract the scalar mean from each of the three matrices and apply the cited simultaneous hollowisation theorem. In the resulting basis the first two means are attained at every vector. All but possibly two of the diagonal entries of E equal $\mu = \text{tr } E/R$. The remaining two entries have sum 2μ and are nonnegative; each is therefore at most 2μ . This argument also covers $R = 2$. $\square$

We also use the following theorem of Marcus, Spielman, and Srivastava [5, Theorem 1.4]. If independent, finitely supported random vectors x_i in a complex Hilbert space satisfy

$$\sum_i \mathbb{E} x_i x_i^* = I, \quad \mathbb{E} \|x_i\|^2 \leq \varepsilon, \quad (6.2)$$

where $\varepsilon > 0$, some outcome satisfies

$$\left\| \sum_i x_i x_i^* \right\| \leq (1 + \sqrt{\varepsilon})^2. \quad (6.3)$$

The same conclusion holds if the covariance sum is merely at most I . Indeed, spectrally decompose its positive semidefinite deficit and split each eigenvalue into finitely many pieces at most ε . Add the corresponding deterministic vectors, apply the theorem, and then discard their positive semidefinite contributions. No upper restriction on ε is required.

6.2 Paving with exact quotas in every group

Lemma 6.2. *Let $D \geq 0$ and let h be a nonnegative integer. Suppose vectors are indexed by finitely many groups of size $R = 2^h$ and satisfy*

$$\sum_i f_i f_i^* \leq I, \quad \|f_i\|^2 \leq D/R.$$

Their indices can be partitioned into R transversals, each containing exactly one index from each group, such that every transversal J satisfies

$$\left\| \sum_{i \in J} f_i f_i^* \right\| \leq \frac{C_D}{R}, \quad C_D = [1 + 2(\sqrt{2} + 1)\sqrt{D}]^2. \quad (6.4)$$

Proof. If $D = 0$ all the vectors vanish, so any transversal partition works. Assume $D > 0$ and write $\delta = D/R$. Consider a node containing equally many indices from each group, an even number per group, whose frame operator is at most LI . Pair the indices within each group. For a pair of vectors a, b , choose independently and uniformly one of the four vectors

$$\sqrt{2}(a \oplus b), \quad \sqrt{2}(a \oplus (-b)), \quad \sqrt{2}(b \oplus a), \quad \sqrt{2}(b \oplus (-a)). \quad (6.5)$$

For the random vector w so obtained, the off-diagonal expected covariance blocks cancel. On summing over the pairs,

$$\sum \mathbb{E} w w^* = \text{diag} \left(\sum_i f_i f_i^*, \sum_i f_i f_i^* \right) \leq LI, \quad \mathbb{E} \|w\|^2 \leq 4\delta.$$

When $L > 0$, apply (6.3) to $w/\sqrt{L}$ with $\varepsilon = 4\delta/L$. Some outcome has

$$\left\| \sum w w^* \right\| \leq (\sqrt{L} + 2\sqrt{\delta})^2.$$

Its two diagonal blocks equal twice the frame operators of two complementary halves, each containing one member of each pair. Consequently both children have frame operator bounded by

$$L_{\text{next}} I, \quad L_{\text{next}} = \frac{1}{2}(\sqrt{L} + 2\sqrt{\delta})^2. \quad (6.6)$$

If $L = 0$, the vectors at the node are zero and can be divided arbitrarily.

Repeat this operation at every node for h levels, starting with $L_0 = 1$. At depth j every node contains $R/2^j$ indices from each original group. Thus the leaves are disjoint transversals covering all indices. Taking square roots in (6.6) and summing the geometric progression gives

$$\sqrt{L_h} \leq 2^{-h/2} + \sqrt{2\delta} \sum_{j=0}^{h-1} 2^{-j/2} = R^{-1/2} + 2(\sqrt{2} + 1)\sqrt{D/R}(1 - R^{-1/2}),$$

which implies (6.4). The construction partitions indices, so repeated or zero vector values cause no ambiguity. $\square$

6.3 Trace-preserving small compressions

Proposition 6.3. *Let $h \geq 1$ and $k \geq 1$ be integers, set $R = 2^h$ and $n = kR$, and suppose $\text{tr } A = 0$ and, for some $\theta \in \mathbb{R}$,*

$$\text{tr} |\text{Re}(e^{i\theta} A)| \leq 2k \|A\|. \quad (6.7)$$

There are R mutually orthogonal rank- k projections P_ℓ summing to I such that

$$\text{tr}(P_\ell A P_\ell) = 0, \quad \|P_\ell A P_\ell\| \leq \frac{319}{R} \|A\|. \quad (6.8)$$

Proof. The case $A = 0$ is immediate. Normalize and rotate so that $\|A\| = 1$ and $A = H + iG$, where H, G are trace-zero Hermitian matrices and

$$E = |H| \leq I, \quad \text{tr } E \leq 2k.$$

Choose an orthonormal eigenbasis e_i of G , with eigenvalues λ_i . Set

$$d_i = e_i^* E e_i, \quad \mu = \frac{\text{tr } E}{kR}.$$

Since $|\lambda_i| \leq 1$, $0 \leq d_i, \mu \leq 1$, the real vectors $(\lambda_i, d_i - \mu)$ sum to zero and have ℓ_∞ norm at most one. Apply Steinitz rearrangement and divide the reordered list into k consecutive groups of size R . Each group sum is a difference of two partial sums. The corresponding orthogonal spectral subspaces V_j therefore satisfy

$$|\text{tr } G_j| \leq 4, \quad \text{tr } E_j \leq R\mu + 4 \leq 6, \quad (6.9)$$

where subscripts denote compression to V_j . The spaces V_j reduce G .

Apply Lemma 6.1 to H_j, G_j, E_j in each V_j . This gives an orthonormal basis $v_{j,a}$ with

$$v_{j,a}^* H v_{j,a} = \text{tr } H_j / R, \quad v_{j,a}^* G v_{j,a} = \text{tr } G_j / R, \quad v_{j,a}^* E v_{j,a} \leq 12/R. \quad (6.10)$$

For $f_{j,a} = E^{1/2} v_{j,a}$ we have

$$\sum_{j,a} f_{j,a} f_{j,a}^* = E \leq I, \quad \|f_{j,a}\|^2 \leq 12/R.$$

Lemma 6.2 with $D = 12$ gives R transversals. Let P_ℓ project onto the span of the original basis vectors in transversal ℓ . These projections are orthogonal, sum to I , and have rank k .

Equation (6.10) implies that the trace of each compression of A is

$$\frac{1}{R} \sum_j (\text{tr } H_j + i \text{tr } G_j) = \frac{\text{tr } A}{R} = 0.$$

Since distinct V_j reduce G , the compression $P_\ell G P_\ell$ is diagonal in its selected basis, and (6.9) gives $\|P_\ell G P_\ell\| \leq 4/R$. Furthermore

$$\|P_\ell E P_\ell\| = \|E^{1/2} P_\ell E^{1/2}\| = \left\| \sum_{(j,a) \in \ell} f_{j,a} f_{j,a}^* \right\| \leq C_{12}/R.$$

The first equality follows from $\|X^* X\| = \|X X^*\|$ for $X = E^{1/2} P_\ell$. Since $-E \leq H \leq E$, order and the Hermitian norm formula imply $\|P_\ell H P_\ell\| \leq C_{12}/R$. Thus

$$\|P_\ell A P_\ell\| \leq (C_{12} + 4)/R < 319/R.$$

For completeness, $\sqrt{2} < 1.415$ and $\sqrt{12} < 3.465$ give $1 + 2(\sqrt{2} + 1)\sqrt{12} < 17.736$, whose square is less than 315. Undo the rotation and normalization to obtain (6.8). $\square$

7 Finite-block assembly and the global induction

Lemma 7.1. *With s, R, b as in (1.2), suppose A is decomposed into $1 \leq r \leq 2R - 1$ nonempty orthogonal blocks and every diagonal block has a representation $A_{ii} = [U_i, V_i]$ of cost $p_i = \|U_i\| \|V_i\|$. Then $A = [B, C]$ in its original dimension, with*

$$\|B\| \|C\| \leq 8s \max_i p_i + b \|A\|. \quad (7.1)$$

Proof. For a nonzero diagonal block, inversely rescale its two factors to make $\|U_i\| = 1/4$ and $\|V_i\| = 4p_i$. For a zero block use two zero factors; this only decreases its cost. Choose distinct centers z_i from the square grid

$$\{x + iy : x, y \in \{-s + \frac{1}{2}, -s + \frac{3}{2}, \dots, s - \frac{1}{2}\}\}.$$

There are $4s^2 = 4R$ centers, with pairwise distance at least one. Define

$$B = \text{diag}(z_i I + U_i), \quad C_{ii} = V_i.$$

Since $|z_i| \leq \sqrt{2}(s - 1/2)$, we have $\|B\| < 2s$. For each $i \neq j$ solve

$$(z_i - z_j)C_{ij} + U_i C_{ij} - C_{ij} U_j = A_{ij}.$$

Lemma 4.1 applies with perturbation norm at most $1/2$, giving $\|C_{ij}\| \leq 2 \|A_{ij}\| \leq 2 \|A\|$. The block equations give $[B, C] = A$, and (1.3) implies

$$\|C\| \leq 4 \max_i p_i + 2(r - 1) \|A\|.$$

Multiply by $2s$ and use $r - 1 \leq 2R$ to obtain (7.1). No normality assumption on any U_i is used. $\square$

Proof of Theorem 1.1. Let $K_H = K_H(t_0)$ be defined by (5.3). We first prove the bound with the internal budget

$$K_0 = \max\{8sK_H + b, 2b\} = \max\{65536K_H + 2^{42}, 2^{43}\}. \quad (7.2)$$

This is a finite absolute constant. In this induction write $K = K_0$; the quantitative calculation below shows $K_0 \leq 2^{3200}$. We argue by strong induction on n . The zero matrix has the required representation, which also covers dimension one. For a nonzero A , put $a = \|A\| > 0$.

First choose an orthonormal basis in which A has zero diagonal. Here is a short justification. The average of the diagonal expectations in any orthonormal basis is zero. By convexity of the numerical range, zero is a vector expectation. Choose a unit vector with this expectation; the compression to its orthogonal complement still has trace zero. Repeat in the smaller dimension. This also follows from [2, Corollary 14].

If $n < R$, use the n zero singleton diagonal blocks in Lemma 7.1; the cost is at most $ba \leq Ka$. If $n \geq R$, write

$$n = kR + q, \quad k \geq 1, \quad 0 \leq q < R.$$

Retain $N = kR$ vectors of the zero-diagonal basis and write A_0 for the compression to their span. The other q vectors are retained as zero singleton diagonal blocks. Then $\text{tr } A_0 = 0$ and $\|A_0\| \leq a$. All off-diagonal interactions remain part of A and will be handled by the assembly lemma.

If $A_0 = 0$, every diagonal block in this decomposition is zero, and the assembly lemma applies directly. Otherwise there are two cases.

If A_0 satisfies $\text{HM}(t_0)$, Proposition 5.3 gives a representation of A_0 of cost at most $K_H \|A_0\| \leq K_H a$. Assembling this block with the q zero singletons gives cost at most $(8sK_H + b)a \leq Ka$. The

condition is tested with A_0 's own norm. When $q = 0$, this step is still a direct application of the high-mass proposition; it does not appeal to induction in dimension n .

If A_0 fails $\text{HM}(t_0)$, some θ satisfies

$$\text{tr} |\text{Re}(e^{i\theta} A_0)| < t_0 N \|A_0\| = 2k \|A_0\|.$$

Proposition 6.3 supplies R trace-zero blocks T_i of dimension $k < n$, with $\|T_i\| \leq (319/R)a$. Strong induction gives commutator representations of these blocks of cost at most $(319K/R)a$. Together with the q zero singletons there are $R + q \leq 2R - 1$ blocks. Applying Lemma 7.1 gives

$$\|B\| \|C\| \leq \left(\frac{8s \cdot 319}{R} K + b \right) a = \left(\frac{2552}{8192} K + b \right) a \leq Ka,$$

because $2552/8192 < 1/2$ and $K \geq 2b$. This closes the induction. $\square$

7.1 Auditing the explicit constant

At $t_0 = 2^{-25}$ the high-mass construction has

$$a_0 = 2^{27}, \quad L = 2^{29}, \quad k_* = L + 1 \leq 2^{30}, \quad \sqrt{L} \leq 2^{15}.$$

The two-shear budgets in (5.3) therefore satisfy

$$u_* \leq 2^{28}, \quad a_{1,*} \leq 2^{83}, \quad 1 + \sqrt{L} a_{1,*} \leq 2^{99}, \quad 1 \leq a_* \leq 2^{281}. \quad (7.3)$$

For clarity we bound the core polynomial directly. Its scale, as defined in Section 3, is

$$\sigma(x) = 2 \max\{(5x + 1)^2 + (2x + 1)(5x + 1) + 14x + 1, 12x + 3\}.$$

For $x \geq 1$, the first expression in the maximum is $35x^2 + 31x + 3 \leq 69x^2$, and the second is at most $15x^2$. Consequently $\sigma(x) \leq 138x^2$ and

$$\begin{aligned} P(x) &= (5x + 2)^4 (\sigma(x) + 3) ((\sigma(x) + 5)2x + 2) \\ &\leq (7x)^4 (141x^2) (288x^3) \leq 2^{29} x^9. \end{aligned}$$

Thus $P(a_*) \leq 2^{2558}$. The pullback through the initial normalization and the two shears costs at most $a_*^2 (a_*/a_0)^2 = a_*^2$. Substitution in (5.3) gives

$$\begin{aligned} K_H(t_0) &\leq (2^{281})^2 (4 \cdot 2^{30} \cdot 2^{2558} + 2(2^{30})^2 \cdot 2^{281}) \\ &= 2^{3152} + 2^{904} \leq 2^{3153}. \end{aligned}$$

The high-case assembly budget is at most $2^{16} 2^{3153} + 2^{42} < 2^{3170}$. In the low case the cost is

$$\left(\frac{319}{1024} K + 2^{42} \right) \|A\| \leq K \|A\| \quad \text{whenever } K \geq 2^{43}.$$

It follows that the internal budget K_0 in (7.2) is at most 2^{3200} , proving the explicit value in Theorem 1.1. All these comparisons are exact inequalities in the Lean proof; no floating-point arithmetic is used.

Remark 7.2 (Scope of the bound). The exponent 3200 is a convenient generous choice, not an optimal exponent. The improvement is that precisely two outside-block shears suffice, independent of the number of outside blocks. The factors need not be normal. Auxiliary doubled spaces occur only in the vector-selection argument; the final commutator representation always acts on the original space.

A Lean correspondence and reproducibility

A.1 The strengthened statement and its norm

The public entry point `CommutatorTheorem.lean` imports the new module `CommutatorTheorem/NoEpsilon/ExplicitBound.lean`. It activates `Matrix.Norms.L2Operator`. The final declaration, with no hidden section assumptions, states

```
commutator_bound_two_pow_3200
(n : Nat) (A : Matrix (Fin n) (Fin n) Complex)
(hA : Matrix.trace A = 0) :
  exists B C : Matrix (Fin n) (Fin n) Complex,
    A = B * C - C * B and
    norm B * norm C <= (2 : Real) ^ 3200 * norm A
```

Here the displayed ASCII notation is a readable transcription; the audit file prints the elaborated Lean type. In particular the constant precedes all choices of dimension and matrix, the trace is unnormalized, and both witnesses have exactly the original matrix type. The theorem also covers $n = 0$.

The declaration with suffix `_euclideanOperatorNorm` uses `Matrix.toEuclideanCLM` in each norm. Its proof is the matrix-norm declaration itself, with no norm-equivalence factor: the norms are definitionally the same Euclidean operator norm. Thus this is not an entrywise or Frobenius norm estimate.

A.2 Mathematical correspondence

Names use namespace `NoEpsilon`. All files are in the directory `CommutatorTheorem/NoEpsilon/`.

Mathematical step	File and declaration
Orthogonal sum estimate	<code>SimultaneousAbsorption.lean:</code> <code>Absorption.orthogonal_sum_norm_sq</code>
Two simultaneous shears	<code>SimultaneousAbsorption.lean:</code> <code>Absorption.eliminate_simultaneously</code>
Core and outside assembly	<code>AbsorptionCore.lean:</code> <code>Absorption.identity_corner_absorption</code>
Diagonal bridge normalization	<code>DiagonalAbsorption.lean:</code> <code>Absorption.diagonal_corner_absorption</code>
High-mass branch	<code>HighMassTheorem.lean:</code> <code>highMass_bounded_commutator</code>
Exact low-mass paving	<code>LowMassTheorem.lean:</code> <code>normalized_low_mass_paving</code>
Same-dimension induction	<code>GlobalInduction.lean:</code> <code>boundedCommutator_of_lowMassBlocksAt</code>
Explicit numerical estimates	<code>ExplicitBound.lean:</code> <code>highMassNormBudget_fixed_le,</code> <code>globalNormBudget_le_explicit</code>
Theorem 1.1	<code>ExplicitBound.lean:</code> <code>commutator_bound_two_pow_3200</code>

The sequential elimination lemmas remain available for compatibility, but the refined main proof uses the simultaneous theorem. Its generic coordinate proof uses orthogonal isometries V_i , the projection $E = \sum_i V_i V_i^*$, and the row $L = \sum_i J_i V_i^*$. It proves $\|E\| \leq 1$ and $\|L\| \leq \sqrt{q}$, then takes $N = Q(L - P^* A E)$ and $M = (\sum_i V_i (V_i^* A' V_i J_i^*)) P^*$. These are precisely the two shears

in Proposition 4.2. It proves $N^2 = M^2 = 0$, exact block identities, trace preservation, and the condition-number and norm bounds, including empty outside slots.

The low-mass result retains the existing proof through Steinitz grouping, simultaneous hollowization, and paired MSS selection. The MSS development passes through stability of positive-semidefinite matrix pencils, the mixed characteristic polynomial barrier bound, and interlacing selection. These are proved inputs, rather than additional assumptions of the final theorem. The final induction explicitly retains leftover zero singletons in the original dimension and tests the normalized retained compression against its own norm.

A.3 Build and trust audit

This revision uses Lean v4.30.0-rc1 and the pinned Mathlib revision in `lake-manifest.json`. The exact Mathlib commit and SHA-256 hashes of the changed proof sources are recorded in `verification/explicit_3200/manifest.json`. The base repository commit is

8560e3605900427b4b8f615e58cd2d488262ad07.

The revision is a local working-tree change, not a newly published commit. The recorded source hashes identify the checked refinement. From the repository root, reproduce the checks with

```
lake build
lake env lean verification/explicit_3200/Audit.lean
```

The audit prints the final types and the transitive axiom sets for the simultaneous elimination, the numerical high-case estimate, and both final norm formulations. The final declarations depend only on

[propext, Classical.choice, Quot.sound].

There is no `sorryAx`, custom mathematical axiom, or native decision axiom in these dependencies. Full build and audit output accompanies the source under `verification/explicit_3200/`. A type audit is included because an axiom list alone would not rule out hidden mathematical parameters.

References

- [1] I. Bárány, *A matrix version of the Steinitz lemma*, arXiv:2308.10102v2 (2024). <https://arxiv.org/abs/2308.10102v2>.
- [2] T. Damm and H. Faßbender, *Simultaneous hollowisation, joint numerical range, and stabilization by noise*, arXiv:1910.08813v2 (2020). <https://arxiv.org/abs/1910.08813v2>.
- [3] W. B. Johnson, N. Ozawa, and G. Schechtman, *A quantitative version of the commutator theorem for zero trace matrices*, arXiv:1202.0986 (2012). <https://arxiv.org/abs/1202.0986>.
- [4] C.-K. Li and N.-S. Sze, *Canonical forms, higher rank numerical ranges, totally isotropic subspaces, and matrix equations*, arXiv:0706.1536 (2007). <https://arxiv.org/abs/0706.1536>.
- [5] A. W. Marcus, D. A. Spielman, and N. Srivastava, *Interlacing families II: Mixed characteristic polynomials and the Kadison–Singer problem*, arXiv:1306.3969v4 (2014). <https://arxiv.org/abs/1306.3969v4>.
- [6] M. Ravichandran and N. Srivastava, *Asymptotically optimal multi-paving*, arXiv:1706.03737v2 (2017). <https://arxiv.org/abs/1706.03737v2>.